Suhas M Shivakumar

Aerospace Engineer

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Summary

Aerospace Engineer with nearly 3 years of experience in composite materials testing, laboratory-scale manufacturing, and simulation-supported material validation. Experienced in specimen preparation, standards-based testing, failure analysis, and laboratory support for material development and process optimization.

Experience

Adithya Industries **Apr 2025 – Apr 2026**

Structural Engineer *Remote – Munich, Germany*

- Supported materials validation and structural assessment activities through FEM-based correlation of test data and simulation results for aerospace-grade composite components.
- Prepared technical reports, validation summaries, and engineering documentation to support structured decision-making and product development reviews.
- Contributed to material qualification workflows involving stress, stiffness, and failure evaluation under defined load cases.

Chair of Carbon Composites (LCC), TUM **July 2023 – Mar 2025**

Research Assistant *Munich, Germany*

- Planned and conducted mechanical testing campaigns on CFRP laminates using UTM, including tensile, compression, flexural, fatigue, and cryogenic tests according to ASTM, DIN, and ISO standards.
- Performed thermo-analytical and material characterization activities including DSC-based resin analysis, optical microscopy, and SEM failure investigations.
- Manufactured composite laminates at laboratory scale using VARTM, AFP, filament winding, and hand lay-up processes, including resin handling and curing workflows.
- Documented test procedures, results, and failure modes while supporting industrial R&D projects on material development and process optimization.

Engineering Projects

3D Woven L-Stringers – Stiffness Improvement | *Composite Manufacturing, Material Modeling, Structural analysis*

- Developed and validated a 3D woven composite L-stringer concept for aerospace structural applications, generating quantitative evidence of up to 35% improved out-of-plane load performance versus traditional 2D laminates.

Education

Master of Science in Aerospace Engineering **Munich, Germany**

Technical University of Munich *Oct 2021 – Mar 2025*

Bachelor of Engineering in Aeronautics **Bengaluru, India**

Nitte Meenakshi Institute of Technology *Jul 2016 – Aug 2020*

Skills

- **Laboratory Testing**: Universal Testing Machine (UTM), tensile/compression/flexural/fatigue testing, specimen preparation, data acquisition, standards-based testing (ASTM, DIN, ISO)
- **Material Characterization**: DSC, optical microscopy, SEM, fracture and failure analysis, thermo-mechanical evaluation, resin curing assessment
- **Composite Processing**: CFRP laminate manufacturing, thermoset resin systems, VARTM, AFP, filament winding, hand lay-up, cure cycle understanding
- **Engineering Tools**: ABAQUS, ANSYS, CATIA V5, Python, MATLAB, MS Office
- **Documentation & Standards**: Technical reporting, test documentation, structured data archiving, safety compliance, laboratory workflow discipline
- **Languages**: English (C1), German (B1 progressing)